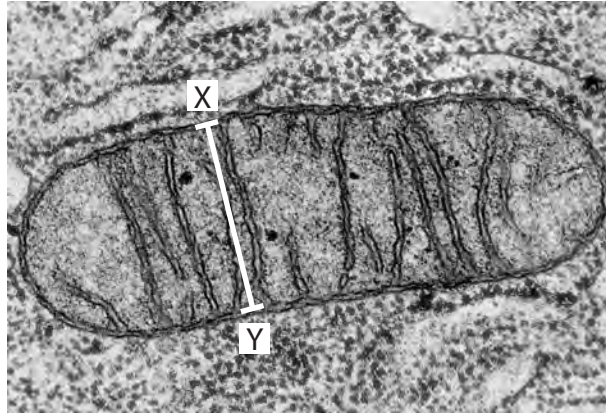


- 1 The photograph shows a transmission electron micrograph of a mitochondrion and surrounding cytoplasm.



(Source: © CNRI / SCIENCE PHOTO LIBRARY)

- (a) The mitochondrion shown in the diagram is 2000 nm in length.

Calculate the width of this mitochondrion between X and Y.

(2)

Answer nm

(b) Describe how ATP is synthesised by oxidative phosphorylation in the mitochondrion.

(3)

(c) Describe how ATP is used to supply energy for biological processes.

(2)

(Total for Question 1 = 7 marks)